

TE Curve Verification Pipeline Official Model Verification Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Verdict

This official closure report accepts the completed wave 4.4 latent-state hysteresis refresh into the canonical TE Curve Verification Pipeline evidence package.

Decision

- track2h_latent_state_hysteresis_registry is closed as a verified exploratory baseline.
- No wave 4.4 candidate is promoted over the accepted direction-parallel leaders.
- The strongest refreshed wave 4.4 aggregate candidate is track2h_l_causal_tcn_offset_residual_global .
- The accepted direction-parallel leaders remain rcim_retuned_GBM19_Fw , periodic_gru_sequence_Bw , and the accepted global neural periodic_gru_sequence_global .
- The closure keeps latent-state / hysteresis-aware modeling as integration evidence for later multi-head work, not as the next standalone promoted family.

Source Package

This official report consolidates these refreshed artifacts:

- metric matrix:
doc/reports/analysis/track2/Track 2 Directional Model Comparison.md ;
- matrix summary:
output/validation_checks/track2_reference_comparison/2026-06-18-15-20-21__track2_full_directional_family_matrix_track2h_latent_state_hysteresis_track2_refresh_2026_06_18/validation_summary.yaml ;
- per-condition metrics:
output/validation_checks/track2_reference_comparison/2026-06-18-15-20-21__track2_full_directional_family_matrix_track2h_latent_state_hysteresis_track2_refresh_2026_06_18/per_condition_metrics.csv ;
- best-model collage report:
doc/reports/analysis/track2/best_model_collage_report/[2026-06-18]/track2_best_model_collage_report.md ;
- multi-model curve comparison report:
doc/reports/analysis/track2/multi_model_curve_comparison_report/[2026-06-18]/track2_multi_model_curve_comparison_report.md ;
- operator launch logs:
output/validation_checks/track2_operator_launch_logs/2026-06-18-15-20-09_track2h_latent_state_hysteresis_track2_refresh_2026_06_18 .

Candidate Refresh

The refresh added 6 candidates from track2h_latent_state_hysteresis_registry into the official 165 - candidate matrix.

Surface	Candidate	Family
global	track2h_l_gru_offset_residual_global	track2h_l_gru_offset_residual
Fw	track2h_l_gru_offset_residual_Fw	track2h_l_gru_offset_residual
Bw	track2h_l_gru_offset_residual_Bw	track2h_l_gru_offset_residual
global	track2h_l_causal_tcn_offset_residual_global	track2h_l_causal_tcn_offset_residual
Fw	track2h_l_causal_tcn_offset_residual_Fw	track2h_l_causal_tcn_offset_residual
Bw	track2h_l_causal_tcn_offset_residual_Bw	track2h_l_causal_tcn_offset_residual

Refreshed Source Leaders

The table ranks the refreshed source by aggregate offline TE Curve Verification Pipeline metrics.

Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
Fw	track2h_1_causal_tcn_offset_residual_Fw	0.003476	0.003939	7.717	13.330
Bw	track2h_1_gru_offset_residual_Bw	0.003542	0.003984	7.736	12.831
global	track2h_1_causal_tcn_offset_residual_global	0.003372	0.003827	7.398	13.454

Interpretation

- global : causal TCN is the best Wave 4.4 aggregate candidate, but it remains behind the accepted global neural baseline and behind the strongest probabilistic Wave 4 series global scalar evidence.
- Fw : causal TCN is the best Wave 4.4 forward candidate, but its 0.003476 deg curve MAE is far behind rcim_retuned_GBM19_Fw at 0.001089 deg .
- Bw : GRU is the best Wave 4.4 backward candidate, but its 0.003542 deg curve MAE is behind periodic_gru_sequence_Bw at 0.002392 deg .

Refreshed Source Leaderboard

Rank	Surface	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
1	global	track2h_1_causal_tcn_offset_residual_global	0.003372	0.003827	7.398	13.454
2	Fw	track2h_1_causal_tcn_offset_residual_Fw	0.003476	0.003939	7.717	13.330
3	Bw	track2h_1_gru_offset_residual_Bw	0.003542	0.003984	7.736	12.831
4	Fw	track2h_1_gru_offset_residual_Fw	0.003549	0.003996	7.873	12.664
5	global	track2h_1_gru_offset_residual_global	0.003591	0.004024	7.896	12.986
6	Bw	track2h_1_causal_tcn_offset_residual_Bw	0.003624	0.004098	7.903	13.135

Current Direction Leaders

These leaders are read from the matrix direction breakdown after the refresh.

Direction	Candidate	MAE [deg]	RMSE [deg]	Mean [%]	P95 [%]
backward	periodic_gru_sequence_Bw	0.002392	0.002639	5.466	14.820
forward	rcim_retuned_GBM19_Fw	0.001089	0.001299	2.372	4.912

The current accepted direction-parallel baselines do not change.

Multi-Index Closure

This refresh provides raw-error and visual-evidence coverage for the `wave 4.4` source. The companion collage and overlay reports show that the latent-state candidates follow the broad TE curve trend, but they still show visible offset and high-frequency mismatch on representative conditions. The available evidence does not justify promotion on raw error, offset / continuity behavior, harmonic / phase fidelity, robustness, or deployment readiness.

The official closure therefore treats `wave 4.4` as useful evidence that causal history can help the `global` scalar surface, while rejecting it as a standalone replacement for the accepted forward, backward, or global TE Curve Verification Pipeline leaders.

Visual Evidence

The same launcher run regenerated the visual companion reports and verified that the refreshed source appears in the visual package.

Source	Collage	Overlay Forward	Overlay Backward
<code>track2h_latent_state_hysteresis_registry</code>	6	2	2

Closeout Decision

`wave 4.4 latent-state hysteresis refresh` is closed as a verified exploratory baseline and is not promoted. Use the latent-state hysteresis evidence as a later integration ingredient when comparing multi-head designs that combine causal state, offset handling, uncertainty, mixture pressure, and structured harmonic residuals.